

Deciphering the relationship between type 2 diabetes and fracture risk: the genetic and observational evidences

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Abstract

The “diabetic bone paradox” suggested that type 2 diabetes (T2D) patients would have higher bone mineral density (BMD) but higher fracture risk than individuals without T2D. In this study, we found that the genetically predicted T2D was associated with higher BMD and lower risk of fracture in both wGRS and two-sample MR analyses. We also identified ten genomic loci shared between T2D and fracture, with the top signal at SNP rs4580892 in the intron of gene *RSPO3*. And the higher expression in adipose subcutaneous and higher protein level in plasma of *RSPO3* were associated with increased risk of T2D, but decreased risk of fracture. In the prospective study, T2D was observed to be associated with higher risk of fracture, but BMI mediated 30.2% of the protective effect. However, when stratified by the risk factors secondary to the disease, we observed that the effect of T2D on the risk of fracture decreased when the number of risk factors secondary to T2D decreased, and the association became non-significant if the T2D patients carried none of the risk factors. In conclusion, the genetically determined T2D might not be associated with higher risk of fracture. And the shared genetic architecture between T2D and fracture suggested a top signal around *RSPO3* gene. The observed effect of T2D on fracture risk decreased if the risk factors secondary to T2D could be eliminated. Therefore, it is important to manage the complications of T2D to prevent the risk of fracture.

eLife assessment

The increased risk of fracture without decreased bone density in type 2 diabetes (T2D), the so-called "diabetic bone paradox", is mainly attributed to the limitation of assessing risk of fracture based on bone density alone in the current practice. Now we have learnt that poor bone quality and increased risk of falling due to concomitant co-morbidities partially explains it. This study presents **useful** findings that clinical risk factors (though not genetic factors) related to T2D are associated with risk of fracture in T2D patients. The new approach of using Mendelian randomization to explain the relationship of two complex conditions is **solid**, and the discovery of 10 loci shared between T2D and fracture risk is intriguing. However, including clinically more relevant risk factors for fracture risk in T2D patients in their observational analysis would have strengthened the study.

Introduction

Type 2 diabetes (T2D), a chronic metabolic disorder characterized by elevated blood glucose levels and increased risk of numerous serious and life-threatening complications, constitutes one of the biggest health problems in the world (Diamond, 2003 [↗](#)). According to Global Burden of Disease (GBD) data, the age-standardized global prevalence of type 2 diabetes was approximately 6.0% in men and 5.0% in women in 2019 (Tinajero and Malik, 2021 [↗](#)). It accounts for more than 100 billion dollars of healthcare costs annually in the United States (Diamond, 2003 [↗](#)). The chronic comorbidities of T2D could develop gradually, and could lead to serious damage to heart, blood vessel, kidney, eye and foot (Teck, 2022 [↗](#)). Other organ systems such as skeletal health could also be influenced by T2D (Lei and Kindler, 2022 [↗](#)).

Osteoporosis is an age-related common disease characterized by low bone mass, and fracture is the ultimate outcome of poor bone health. Our previous studies have suggested that bone mass and fracture could be influenced by many modifiable or non-modifiable factors (Zhu and Zheng, 2021 [↗](#)), such as body weight (Zhu et al., 2022 [↗](#)), sleep behavior (Qian et al., 2021 [↗](#)), inflammatory disease (Xia et al., 2020 [↗](#)), birth weight (Xia et al., 2022 [↗](#)) and genetic factors (Zhu et al., 2021 [↗](#)). T2D is also considered to be a major factor that could affect bone health, it seems that T2D patients would have higher bone mineral density (BMD) and higher fracture risk than individuals without T2D (Khosla et al., 2021 [↗](#)). This is the so called "diabetic bone paradox". For example, in an Italian nationwide study of 59,950 women of whom 5.2% had diabetes (predominantly type 2 diabetes), it noted an association between diabetes and any fracture (OR 1.3, 95% CI 1.1–1.4, and OR 1.3, 95% CI 1.2–1.5, for vertebral or hip fractures and non-vertebral, non-hip fractures, respectively) (Adami et al., 2020 [↗](#)). Interestingly, the prevalence of vertebral or hip fracture was higher in participants with diabetes but without obesity (OR 1.9, 95% CI 1.7–2.1) than in patients with obesity and diabetes (OR 1.5, 95% CI 1.3–1.8), suggesting that obesity might be partially protective against vertebral or hip fractures in type 2 diabetes (Adami et al., 2020 [↗](#)).

However, a recent comparative cohort analysis using routinely collected UK primary care records data from the Health Improvement Network (including 174,244 individuals with incident type 2 diabetes and 747,290 without diabetes) found no evidence to suggest a higher risk of fracture in type 2 diabetes patients, specifically, the risk of having at least one fracture was estimated to be 6% lower for females and 3% lower for males in the type 2 diabetes cohort than for females and males without diabetes (Davie et al., 2021 [↗](#)). Lower fracture risk was also observed in the type 2 diabetes patients compared to those without the disease in the age group great than 85 years (Davie et al., 2021 [↗](#)). Another large-scale cohort study showed that type 2 diabetes could only

explain less than 0.1% of the fracture risk (Axelsson et al., 2023), and if the T2D patients with risk factors (such as low BMI, long diabetes duration, insulin treatment, and low physical activity) were excluded, T2D patients would have lower fracture risk than their matched controls (Axelsson et al., 2023). In a prospective study to examine the relationship between BMD and fracture in older adults with type 2 diabetes, it was reported that femoral neck BMD T score and FRAX score were both associated with fracture risk in individuals with type 2 diabetes, suggesting that BMD is still a useful clinical predictor for the evaluation of fracture risk in type 2 diabetes patients (Schwartz et al., 2011).

As the pathophysiology of fracture is more complicated than the BMD trait, and while there were some explanations for the “diabetic bone paradox” (Osorio, 2011), the integrated analyses with genetic data for the diseases could provide an alternative approach to alleviate the bias of the unknown confounding factors (Davey Smith and Hemani, 2014). Therefore, in this study, we firstly performed weighted genetic risk score (wGRS) regression analysis to assess relationship between the genetically predicted T2D and fracture with genetic summary data and individual genotype data in UK biobank. The two-sample mendelian randomization (MR) approach was used as an independent validation analysis. We applied the MiXeR method and conditional/conjunctional false discovery rate (ccFDR) approach to identify the shared genetic components between the traits. Finally, within the UK biobank dataset, the stratified cox regression analyses were applied to explore the association between T2D and fracture risk by including different number of the risk factors secondary to T2D. As complement, the relationship between T2D and BMD was also investigated.

Results

The genetically predicted type 2 diabetes and fracture

We first assessed the relationship of genetically predicted T2D and fracture in UK biobank dataset with the weighted genetic risk score (wGRS) analysis. Within the 294,571 UK biobank samples (see Methods), we constructed the weighted genetic risk score (wGRS) for the individuals in the UK Biobank with the 404 SNPs, which were independently associated with type 2 diabetes (Supplementary Table 1). The wGRS of the 404 SNPs were strongly associated with type 2 diabetes in UK Biobank data ($OR=1.6$, $P<2.0\times 10^{-16}$), suggesting that the instruments were powerful for the MR analysis. When we regressed the observed fracture on the wGRS, we found that the genetically determined type 2 diabetes was associated with lower risk of fracture ($OR=0.982$, $95\%CI=0.975-0.989$, $P=0.006$) (adjusting for reference age, sex, BMI, physical activity, fall history, HbA1c and medication treatments) (Fig. 1a). Similarly, the genetically determined type 2 diabetes was associated with higher BMD ($\beta=0.022$, $95\%CI=0.018-0.026$, $P<2.0\times 10^{-16}$) (Fig. 1b).

We also performed the two-sample MR analyses with fracture GWAS summary data (Trajanoska et al., 2018) which is independent of UK Biobank samples. The inverse variance weighting (IVW) method showed a causal effect of genetically predicted type 2 diabetes on low fracture risk ($OR=0.965$, $95\%CI=0.943-0.988$, $P=0.003$) (Fig. 1a) using 298 SNPs as the instruments (Supplementary Table 2). This causal relationship was also significant in simple median test ($OR=0.967$, $95\%CI=0.936-0.997$, $P=0.033$) (Fig. 1a). There was heterogeneity in IVW results ($Q' p < 0.05$), when we excluded pleiotropic variants using restrictive MR pleiotropy residual sum and outlier test (MR-PRESSO) method, the causal association was still detected between type 2 diabetes and fracture ($OR=0.967$, $95\%CI=0.945-0.989$, $P=0.004$) (Fig. 1a). Moreover, the MR-egger regression also suggested an inverse association between type 2 diabetes and fracture ($OR=0.9666$, $95\%CI=0.9497-0.9828$, $P=0.0002$) (Fig. 1c). The individual effect of the SNPs for type 2 diabetes on fracture was corrected by the false discovery rate (<0.05) (Benjamini and Hochberg, 1995), two of 298 lead SNPs (including rs4580892 near *RSPO3*) of type 2 diabetes remained as potential regions which would also have effect on fracture (Fig. 1c and Supplementary Table 2). Similarly, with

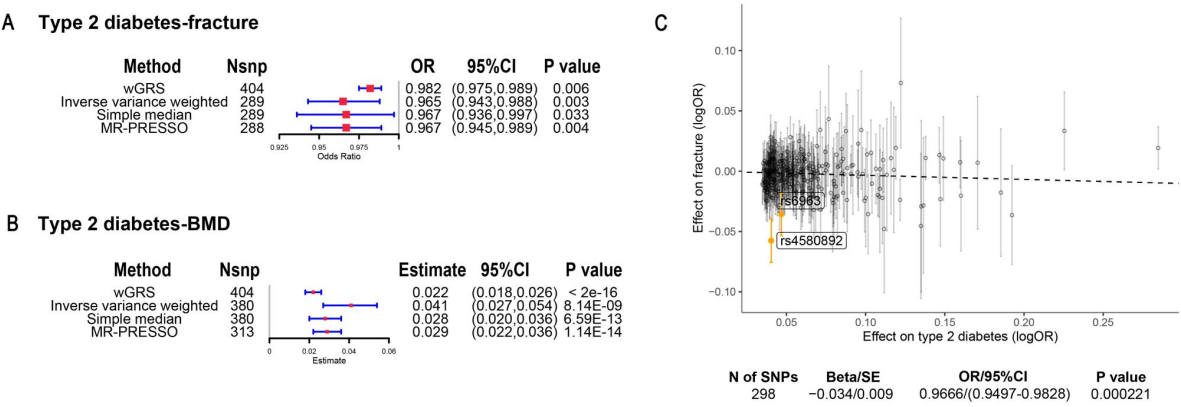


Fig. 1

The association of genetically predicted type 2 diabetes with fracture and BMD by using different MR methods. (a) the genetically predicted type 2 diabetes and fracture. (b) the genetically predicted type 2 diabetes and BMD. (c) visualized the association of lead SNPs for type 2 diabetes with the risk of fracture. Abbreviations: wGRS, weighted genetic risk score; MR-PRESSO, MR pleiotropy residual sum and outlier.

BMD GWAS summary data (Morris et al., 2019), IVW ($\beta=0.041$, 95%CI=0.027-0.054, $P=8.14\times10^{-9}$), simple median ($\beta=0.028$, 95%CI=0.020-0.036, $P=6.59\times10^{-13}$) and MR-PRESSO ($\beta=0.029$, 95%CI=0.022-0.036, $P=1.14\times10^{-14}$) all showed a causal association between type 2 diabetes and BMD (Fig. 1b and Supplementary Table 3).

The distinct signal shared by type 2 diabetes and fracture

Leveraging the genetic summary datasets, we first evaluated the genetic correlation among the traits and diseases by using LDSC (Bulik-Sullivan et al., 2015). It is found that the genetic correlation between type 2 diabetes and fracture was not significant, but with inverse direction ($r_g=-0.0114$) (Supplementary Table 4). Instead, we used MiXeR (Frei et al., 2019) to evaluate the polygenic overlap irrespective of genetic correlation between T2D and fracture. As represented in Venn diagrams of shared and unique polygenic components (Fig. 2a), the MiXeR analysis suggested that type 2 diabetes and fracture exhibited polygenic overlap, sharing 428 causal variants, in other words, 18% of variants (428 of 2370) associated with type 2 diabetes might contribute to the risk of fracture (Dice coefficient=25.25%), and genetic correlation was observed ($r_g=-0.086$) (Fig. 2a and Supplementary Table 5). Only 39% of shared variants between type 2 diabetes and fracture showed concordant direction of association, and the correlation of effect sizes within the shared polygenic component was negative ($\rho_\beta=-0.336$) (Supplementary Table 5).

We used the conditional/conjunctional false discovery rate (ccFDR) approach (Andreassen et al., 2013) to identify specific shared loci between type 2 diabetes and fracture from the GWAS summary statistics. The stratified conditional QQ plot was utilized to visualize the enrichment of association with fracture across varying significance thresholds for type 2 diabetes. We observed leftward deflected from the expected null line in QQ plot, which suggested the existence of polygenic overlap between type 2 diabetes and fracture (Fig. 2b). The conjunctional false discovery rate (conjFDR) analysis identified 10 genomic loci shared between type 2 diabetes and fracture (Fig. 2c and Supplementary Table 6), with the top SNP rs4580892 in the intron of gene *RSPO3* (conjFDR=1.45E-05). The shared loci showed mixed directions of allelic associations, with 7 of 10 shared loci had inverse direction of effect between type 2 diabetes and fracture (Supplementary Table 6). We found that the locus approximately 250kb upstream and downstream of the gene *RSPO3* (hg19, chr6: 127189749-127689749) possessed many significant SNPs associated with type 2 diabetes and fracture, with the nearest gene *RSPO3* (Fig. 3a and 3b). The top SNP rs4580892 had inverse direction of effect between type 2 diabetes and fracture, rs4580892_T allele was associated with increased type 2 diabetes risk (OR=1.041227, $P=8.46\times10^{-9}$) (Fig. 3a) and decreased fracture risk (OR=0.944, $P=3.68\times10^{-10}$) (Fig. 3b).

Further, after merging the eQTL summary data of *RSPO3* in adipose subcutaneous with the summary data of type 2 diabetes and fracture, rs72959041 and rs1936806 were the top cis-eQTL for type 2 diabetes and fracture in GTEx database, respectively. By applying the SMR method (Zhu et al., 2016), we found that the higher expression of *RSPO3* (ENSG00000146374) in adipose subcutaneous would be associated with increased the risk of type 2 diabetes (OR=1.102, 95%CI=1.031-1.179, $P=0.004$), but decreased the risk of fracture (OR=0.695, 95%CI=0.566-0.854, $P=0.0005$) (Fig. 3c and Supplementary Table 7). Interestingly, in adipose subcutaneous, higher expression of *RSPO3* was associated with higher waist circumference ($\beta=0.090$, 95%CI= 0.029-0.151, $P=0.004$) and higher waist-hip ratio ($\beta=0.204$, 95%CI=0.095-0.313, $P=0.0002$) (Fig. 3d and Supplementary Table 7). The association between the expression of *RSPO3* and BMI was not significant, but the direction is the same as the waist circumference (Fig. 3d and Supplementary Table 7). Moreover, to estimate the impact of *RSPO3* protein level on type 2 diabetes and fracture risk, we used the top SNP at rs4580892, a cis-pQTL for circulating *RSPO3* ($P=2.34\times10^{-11}$) identified by Sun et al in an independent dataset (Sun et al., 2018), to instrument the circulating protein level of *RSPO3*. The MR analyses indicated that increased circulating *RSPO3* was strongly associated with increased risk of type 2 diabetes (OR=1.24, 95%CI=1.16-1.34, $P=7.86\times10^{-9}$), but reduced fracture risk (OR=0.73, 95%CI=0.66-0.81, $P=4.1\times10^{-10}$) (Fig. 3e).

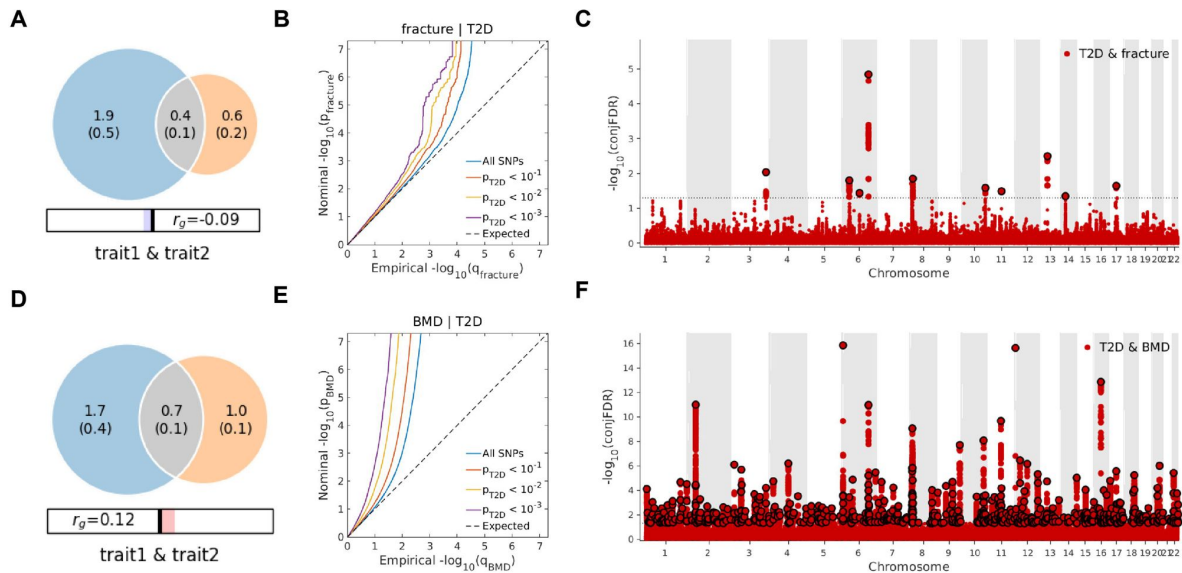


Fig. 2

The venn diagrams, conditional Q-Q plots and genetic variants jointly associated with type 2 diabetes and fracture/BMD at conjunctive false discovery rate (conjFDR) < 0.05. (a) the shared number of variants between type 2 diabetes and fracture. (b) the conditional Q-Q plot of fracture given association with type 2 diabetes at the level of $p \leq 0.1$, $p \leq 0.01$, $p \leq 0.001$. (c) the shared genetic loci between type 2 diabetes and fracture. (d) the shared number of variants between type 2 diabetes and BMD. (e) the conditional Q-Q plot of BMD given association with type 2 diabetes at the level of $p \leq 0.1$, $p \leq 0.01$, $p \leq 0.001$. (f) the shared genetic loci between type 2 diabetes and BMD.

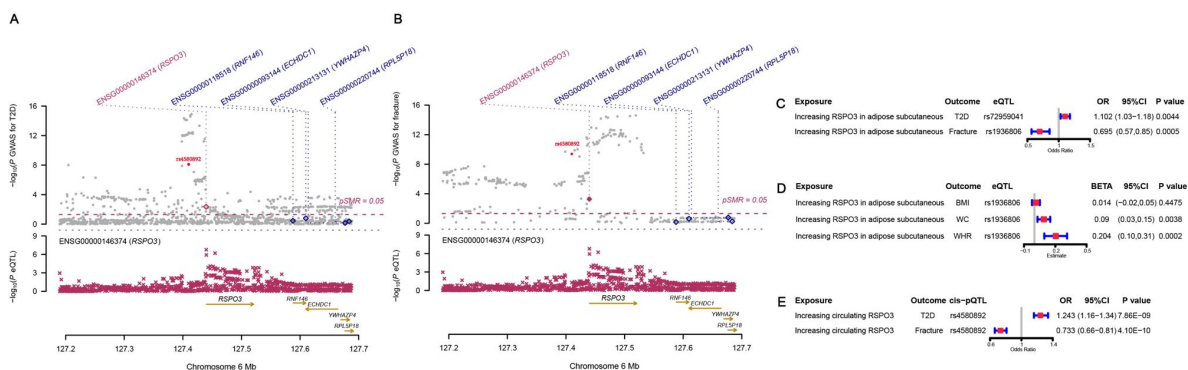


Fig. 3

The distinct signal (*RSP03*) shared by type 2 diabetes and fracture. (a) the regional plot of the association of type 2 diabetes and *RSP03* gene expression (adipose subcutaneous) within hg19: chr6:127189749-127689749 (*RSP03* gene region ± 250 kb window). (b) the regional plot of the association of fracture and *RSP03* gene expression (adipose subcutaneous) in the same region. (c) the association of genetically predicted *RSP03* gene expression with type 2 diabetes and fracture risk. (d) the association of genetically predicted *RSP03* gene expression with BMI, waist circumference and waist-hip ratio. (e) the association of genetically predicted circulating *RSP03* with type 2 diabetes and fracture risk.

Not surprisingly, type 2 diabetes showed significant positive genetic correlation with BMD ($r_g=0.0923$, $P=2.50\times 10^{-6}$) (Supplementary Table 4). The MiXeR analysis suggested that 29% of variants (691 of 2370) associated with type 2 diabetes might contribute to BMD (Dice coefficient=33.67%) (Fig. 2d and Supplementary Table 5). The leftward deflected from the expected null line in QQ plot suggested the existence of polygenic overlap between type 2 diabetes and BMD (Fig. 2e), and the conjFDR analysis identified 661 genomic loci shared between type 2 diabetes and BMD, and 449 of 661 loci (68%) had concordant associations between type 2 diabetes and BMD (Fig. 2f and Supplementary Table 8).

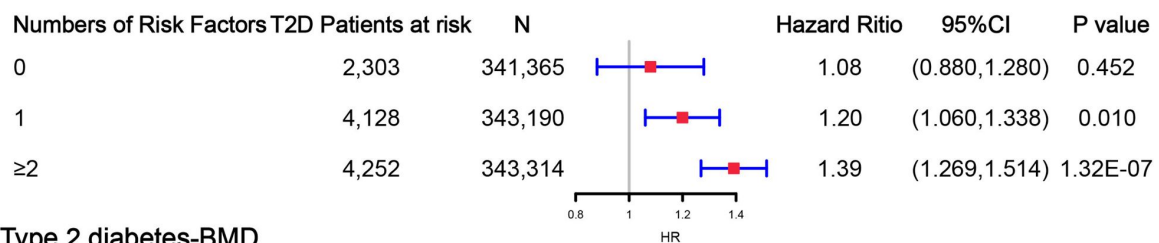
Observed relationship between type 2 diabetes and fracture

Within the 352,879 UK Biobank participants, 13,817 (3.92%) developed type 2 diabetes during 2006 and 2015, with the mean duration of type 2 diabetes 8.34 years (Supplementary Table 9). Compared to those without diabetes, the participants with type 2 diabetes were older (63.20 vs. 60.55, $P<2.2\times 10^{-16}$), and more likely to be men and smokers, and had a higher BMI (32.07 vs. 27.08, $P<2.2\times 10^{-16}$) (Supplementary Table 9). We identified 16,147 (4.6%) participants with fracture within the 352,879 UK Biobank participants (Supplementary Table 9).

Although we found that genetically predicted type 2 diabetes might not be associated with risk of fracture, we observed a higher risk of fracture in the type 2 diabetes patients in the cox proportional hazards regression after adjusted for the reference age, sex, BMI, physical activity, fall history, HbA1c and medication treatments (HR=1.527, 95%CI 1.385-1.685, $P<2\times 10^{-16}$) (Supplementary Table 10). And the average causal mediation effect (ACME) by BMI was protective with 30.2% of the intermediary effect, respectively (BMI: indirect effect=-0.003, $P<2\times 10^{-16}$) (Supplementary Table 11). Inspired from the MR analyses that genetically determined type 2 diabetes might not be a risk factor for fracture, we conducted stratified analyses based on the five risk factors secondary to the diseases, such as BMI $\leq 25\text{kg/m}^2$, no physical activity, falls in the last year, HbA1c $\geq 47.5\text{mmol/mol}$ and antidiabetic medication treatments. Within the 14,860 individuals with type 2 diabetes, 2,303 patients carried none of the risk factors, 4,128 patients accompanied with one of the risk factors, and 4,252 patients carried at least two risk factors (Supplementary Table 12). We performed stratified cox regression analysis and found that type 2 diabetes with at least two risk factors were associated with an increase of fracture risk (HR=1.39, 95%CI 1.269-1.514, $P=1.32\times 10^{-7}$) (Fig. 4). It is interesting to note that including only one risk factor would attenuate the risk effect of type 2 diabetes on fracture risk (HR=1.20, 95%CI=1.060-1.338, $P=0.010$). Furthermore, the association between type 2 diabetes and fracture was not significant ($P=0.452$) when analyzing the type 2 diabetes without risk factors (HR=1.08, 95%CI 0.880-1.280, $N=2,303$) (Fig. 4). These results suggested that the risk factors secondary to type 2 diabetes might contribute to the risk of fracture instead of the disease itself.

We also observed that participants with diabetes, despite they were older, had a significantly higher BMD than subjects without diabetes (0.57 vs. 0.54, $P<2.2\times 10^{-16}$) (Supplementary Table 9). In the multivariable linear regression analysis, the type 2 diabetes was found to be associated with increased BMD in same model adjusted for age, sex, BMI, physical activity, fall history, HbA1c and medication treatments ($\beta=0.00957$, $P=1.35\times 10^{-10}$) (Supplementary Table 10). We examined the relationship between type 2 diabetes and BMD in subgroups with varying numbers of risk factors. We observed that the effect of type 2 diabetes on BMD decreased when the number of risk factors for type 2 diabetes increased (none risk factor: $\beta=0.023$, $P<2\times 10^{-16}$; one risk factor: $\beta=0.020$, $P<2\times 10^{-16}$; at least two risk factors: $\beta=0.017$, $P<2\times 10^{-16}$) (Fig. 4).

A Type 2 diabetes-fracture



B Type 2 diabetes-BMD

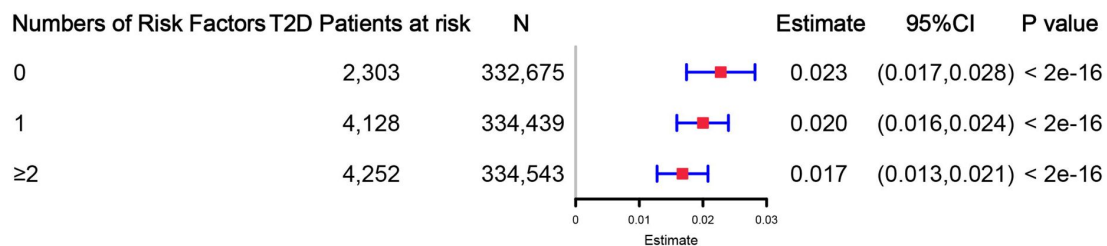


Fig. 4

The stratified analyses between type 2 diabetes, fracture and BMD based on the five risk factors secondary to the diseases adjusted for the age and sex. (a) the relationship between type 2 diabetes and fracture. (b) the relationship between type 2 diabetes and BMD. The 5 risk factors were: (1) $\text{BMI} \leq 25 \text{ kg/m}^2$; (2) no physical activity; (3) falls in the last year; (4) $\text{HbA1c} \geq 7.5 \text{ mmol/mol}$; (5) antidiabetic medication treatments.

Discussion

By leveraging the genetic datasets, we found that the genetically predicted type 2 diabetes was associated with higher BMD and lower risk of fracture in both one-sample MR (with 404 IVs) and two-sample MR (with 298 IVs). We also identified ten genomic loci shared between fracture and type 2 diabetes, with the top signal at SNP rs4580892 in the intron of gene *RSPO3*. And the higher expression of *RSPO3* in adipose subcutaneous was associated with increased the risk of type 2 diabetes, but decreased the risk of fracture. Similarly, the increased circulating *RSPO3* was strongly associated with increased risk of type 2 diabetes, but reduced fracture risk. In the prospective study, type 2 diabetes was observed to be associated with higher risk of fracture, but BMI mediated 30.2% of the protective effect. However, when stratified by the risk factors secondary to the disease, we observed that the effect of type 2 diabetes on the risk of fracture decreased when the number of risk factors secondary to type 2 diabetes decreased, and the association became not significant if the type 2 diabetes patients carried none of the risk factors. The “diabetic bone paradox” suggested that T2D patients would have higher bone mineral density (BMD) but higher fracture risk than individuals without T2D. As reviewed previously, heterogeneity could exist from study to study, and conflicting observational findings were reported (Khosla et al., 2021 [↗](#)). Mendelian randomization (MR) could be an alternative approach to infer the relationship between exposure and outcome, as this method exploits the idea that genotypes are distributed randomly at conception, facilitating their use as instrumental variables (IV) to alleviate the bias of the unknown confounding factors (Davey Smith and Hemani, 2014 [↗](#); Zhao et al., 2019 [↗](#)). Trajanoska K et al (Trajanoska et al., 2018 [↗](#)) assessed the effect of 15 selected clinical risk factors on the risk of fracture by using two-sample MR analysis, and reported non-significant relationship between type 2 diabetes and fracture risk, but the direction of effect was negative (OR=0.99). However, only 38 SNPs were extracted as instruments from a GWAS published in 2012 (Morris et al., 2012 [↗](#)). In the present study, we extracted 298 T2D-associated independent SNPs from Mahajan A et al (Mahajan et al., 2022 [↗](#)), which is the largest-scale GWAS meta-analysis to date published in 2022, as the IVs in two-sample MR analysis. In addition, we also calculated the wGRS with 404 T2D-associated independent SNPs in the UK Biobank dataset, and performed regression analysis of wGRS of type 2 diabetes on the fracture risk (one-sample MR). We found that the genetically predicted type 2 diabetes was associated with lower risk of fracture in both one-sample and two-sample MR analyses. To be note, two-sample MR results could be served as an independent replication to the one-sample MR results, because the effects of the outcome (fracture risk) for two-sample MR were derived Trajanoska K et al (Trajanoska et al., 2018 [↗](#)), while the one-sample MR used the UK Biobank dataset, the study samples had no overlap. Further, the MR analysis in the present study suggested that the genetically predicted type 2 diabetes was associated with higher BMD. That is to say, by alleviating the bias of the unknown confounding factors through MR analysis, the genetically predicted type 2 diabetes did not show this bone paradox.

The genetic correlation between type 2 diabetes and fracture estimated by LDSC (Bulik-Sullivan et al., 2015 [↗](#)) was not significant. It is hard for LDSC to identify the genetic pleiotropy with mixed-effect directions, which is what usually happens between two complex traits. Therefore, in this study, we employed the MiXeR method (Frei et al., 2019 [↗](#)), which could identify the unique and shared polygenic SNPs between two traits regardless of genetic correlation. The MiXeR analysis suggested that type 2 diabetes and fracture exhibited polygenic overlap, and 61% of shared variants between type 2 diabetes and fracture showed discordant direction of association, and the correlation of effect sizes within the shared polygenic component was negative, suggesting an inverse genetic relationship between type 2 diabetes and fracture. Additionally, the conditional/conjunctional false discovery rate analysis (Andreassen et al., 2013 [↗](#)) suggested a top locus at 6q22 (SNP rs4580892) jointly associated with type 2 diabetes and fracture. This SNP is an intronic variant in gene *RSPO3* (ENSG00000146374). We found that higher expression of *RSPO3* in adipose subcutaneous would be associated with increased risk of type 2 diabetes, but decreased risk of fracture. *RSPO3* is a known WNT-signaling modulator (Baron and Kneissel, 2013 [↗](#); Lerner

and Ohlsson, 2015 [↗](#)), which could bind to LRP5/6 to enhance the activity of osteoblast (Richards et al., 2012 [↗](#)). Karin et al demonstrated that *RSPO3* is expressed in osteoprogenitor cells and osteoblasts, and that osteoblast-derived *RSPO3* is the principal source of *RSPO3* in bone and is an important regulator of vertebral trabecular bone mass and bone strength in adult mice (Nilsson et al., 2021 [↗](#)). Interestingly, we also found that higher expression of *RSPO3* was associated with higher waist circumference and higher waist-hip ratio. It was reported that *RSPO3* could impact body fat distribution (Loh et al., 2020 [↗](#)), and fat distribution is an independent predictor of type 2 diabetes (Manolopoulos et al., 2010 [↗](#)). Therefore, we speculated that the different role of the shared genetic components between bone metabolism and type 2 diabetes might provide one possible explanation to the inverse association pattern, and obese tendency might mediate this pattern. In fact, in UK Biobank, the participants with type 2 diabetes had a higher BMI compared to those without diabetes (31.74 vs. 27.05, $P < 2.2 \times 10^{-16}$), and BMI mediated 30.2% of the intermediary effect between type 2 diabetes and fracture in our study.

There are many secondary factors associated with type 2 diabetes might contribute to fracture risk. Our previous study suggested that keeping moderate-high BMI (overweight) might be of benefit to old people in terms of fracture risk (Zhu et al., 2022 [↗](#)), and an intensive lifestyle intervention, such as weight loss, in T2D patients might increase fracture risk (Johnson et al., 2017 [↗](#)). The hyperglycemia could cause osteocyte senescence and premature programmed cell death, leading to decreased ability to sense and respond to mechanical stimuli such as oscillatory shear stress, ultimately contributing to skeletal fragility (Eckhardt et al., 2020 [↗](#)). Another major and complicated factor that might influence the risk of fracture in T2D patients is the use of diabetes mellitus medications. For example, the use of insulin (Napoli et al., 2014 [↗](#)) or thiazolidinediones (Zhu et al., 2014 [↗](#)) was reported to be associated with an increased risk of fracture. In addition, the risk of falls, which might be triggered by some diabetic complications such as visual impairment and peripheral neuropathy, was suggested to increase in the patients with T2D (Schwartz et al., 2002 [↗](#)). Besides, low physical activity was identified as one of the most important independent diabetes-related risk factors for fracture through Gradient Boosting Machines (Axelsson et al., 2023 [↗](#)). Therefore, in this study, we stratified the T2D patients with five secondary risk factors ($BMI \leq 25 \text{ kg/m}^2$, no physical activity, falls in the last year, $HbA1c \geq 47.5 \text{ mmol/mol}$ and antidiabetic medication treatment), and found that the effect of type 2 diabetes on the risk of fracture decreased when the risk factors secondary to type 2 diabetes decreased, and the association became not significant if the type 2 diabetes patients carried none of the risk factors. That is to say, the diabetic bone paradox might not exist if the secondary risk factors of type 2 diabetes were eliminated. In a recent large-scale cohort study, four factors (duration of T2D, low physical activity, BMI, and insulin treatment) were identified as the important risk factors for fracture among the T2D patients, and the patients without the risk factors had lower fracture risk than their matched controls (Axelsson et al., 2023 [↗](#)). One previous study using the UK primary care data found no evidence to suggest a higher risk of fracture in type 2 diabetes patients (Davie et al., 2021 [↗](#)) and reported that significantly lower fracture risk was observed for overweight individuals ($BMI 25\text{-}30 \text{ kg/m}^2$) in type 2 diabetes than their counterparts without type 2 diabetes (Davie et al., 2021 [↗](#)).

In summary, we found that the genetically determined type 2 diabetes might not be associated with higher risk of fracture. And the shared genetic architecture between type 2 diabetes and fracture suggested a top signal near *RSPO3* gene. In addition, the stratified prospective regression analysis suggested that the effect of type 2 diabetes on the risk of fracture decreased if risk factors secondary to type 2 diabetes could be eliminated. These genetic and observational evidences prevailed us to hypothesize that the risk factors secondary to type 2 diabetes might contribute more to the risk of fracture than the disease per se. And it is important to manage the complications of type 2 diabetes to prevent the risk of fracture.

Methods

Study participants and weighted genetic risk score (wGRS) analysis

The UK Biobank data, the application 41376 as we used before (Bai et al., 2020), was applied in our study under a prospective design. We identified the individuals with T2D and fracture using the ICD codes and self-report status. The detail information on the field ID and codes for data extraction from UK Biobank was listed in Supplementary Table 13. We excluded participants if they were identified as follows: 1) ethnically identified as non-European ($n=30,481$); 2) diagnosed as type 1 diabetes ($n=4,455$); 3) diagnosed with diseases associated with bone loss ($n=21,560$); 4) diagnosed with secondary fracture ($n=7,222$) (Supplementary Table 14). For the remaining 439,982 samples, we further excluded 145,411 participants with relatedness (kinship) with others in the wGRS analysis (294,571 participants left) (Supplementary Fig. 1).

The summary-statistic data for the type 2 diabetes were obtained from a very recent GWAS consisting of 80,154 individuals with type 2 diabetes and 853,816 controls in European population (with 10,454,875 SNPs) (Mahajan et al., 2022). We drew a set of independent genetic variants with genome-wide significance ($P < 5 \times 10^{-8}$) from the type 2 diabetes summary-statistic data by LD clumping based on $r^2 < 0.1$ in 500 kb window to serve as instrumental variables ($N=404$) (Supplementary Table 1). We constructed the weighted genetic risk score (wGRS) for the individuals in the UK biobank (294,571 samples with genotypes) as a linear combination of the selected SNPs weighted by their β coefficients on type 2 diabetes: $wGRS = \beta_1 \times SNP_1 + \beta_2 \times SNP_2 + \dots + \beta_n \times SNP_n$. n is the number of instrumental variables (here $N=404$ after LD clumping based on $r^2 < 0.1$ in 500 kb window). Next, cox proportional hazards regression and linear regression analyses were performed to analyze the association between the wGRS and fracture/BMD adjusted age, sex, BMI, physical activity, fall history, HbA1c and medication treatments.

Two-sample Mendelian Randomization (MR) analyses

To validate the wGRS results, we also performed the two-sample MR analyses that is independent of UK Biobank samples. The summary-statistic data for fracture (a discovery set of 37,857 fracture cases and 227,116 controls with 2,539,800 SNPs) (Trajanoska et al., 2018) and BMD (426,824 samples and 13,753,401 SNPs) (Morris et al., 2019) were extracted from the GEFOS consortium (<http://www.gefos.org/>), while the summary data for type 2 diabetes (Mahajan et al., 2022) is the same as used in wGRS analysis.

We used the inverse variance weighting (IVW) (Burgess et al., 2013), simple median and MR-PRESSO (Verbanck et al., 2018) approaches in two-sample MR analyses. For the outcome of fracture, we merged the two summary datasets for T2D and fracture (Mahajan et al., 2022; Trajanoska et al., 2018) and got 2,479,475 overlapping SNPs, of which 6,946 SNPs were genome-wide significant for type 2 diabetes. After LD clumping based on $r^2 < 0.1$ in 500 kb window, 298 independent genetic variants were left (Supplementary Table 2). Similarly, we got 9,204,694 overlapping SNPs for type 2 diabetes and BMD, and 389 independent genetic variants were left after LD clumping (Supplementary Table 3). After harmonizing the effects so that they reflect the same allele, 289 (for fracture) and 380 (for BMD) SNPs were finally used in the IVW and simple median MR analysis. Because the presence of horizontal pleiotropy could bias the MR estimates, we additionally used the MR-PRESSO. The two-sample MR analyses were conducted in R version 4.0.2 using TwoSampleMR (Hemani et al., 2018), MendelianRandomization (Yavorska and Burgess, 2017) and MR-PRESSO (Verbanck et al., 2018) packages. Moreover, we regressed the effects of these 298 SNPs of both traits to highlight the overall effect of T2D on fracture with 'grs.summary' function in the R package 'gtx' (<http://www2.uaem.mx/r-mirror/web/packages/gtx/gtx.pdf>).

Infer the shared genetics

With the summary-statistic GWAS data of type 2 diabetes (Mahajan et al., 2022 [DOI](#)), BMD (Morris et al., 2019 [DOI](#)) and fracture (Trajanoska et al., 2018 [DOI](#)), we performed genome-wide genetic correlation analysis between type 2 diabetes and fracture/BMD by using linkage disequilibrium score regression (LDSC) (Bulik-Sullivan et al., 2015 [DOI](#); Bulik-Sullivan et al., 2015 [DOI](#)) which estimates the degree of shared genetic factors between two traits. We used MiXeR (<https://github.com/precimed/mixer> [DOI](#), version 1.2.0) to quantify polygenic overlap (e.g. how many unique and shared polygenic SNPs for type 2 diabetes and fracture) irrespective of genetic correlation (Frei et al., 2019 [DOI](#)). MiXeR models additive genetic effects as a mixture of four components, representing null SNPs in both traits (π_0); SNPs with a specific effect on the first and on the second trait (π_1 and π_2 , respectively); and SNPs with non-zero effect on both traits (π_{12}). The dice coefficient of two traits was estimated as $\frac{2\pi_{12}}{\pi_1 + \pi_2 + 2\pi_{12}}$ (Frei et al., 2019 [DOI](#)). We constructed conditional quantile-

quantile (QQ) plots which reveals the distribution of P values for fracture/BMD conditioning on the significance of association with type 2 diabetes at the level of $P < 0.1$, $P < 0.01$ and $P < 0.001$ to visualize polygenic enrichment (Schwartzman and Lin, 2011 [DOI](#)). We used the conditional/conjunctive false discovery rate (ccFDR) approach (<https://github.com/precimed/pleiofdr> [DOI](#)) to identify the specific shared loci (Andreassen et al., 2013 [DOI](#)). We used the conditional false discovery rate (condFDR) to detect SNPs associated with fracture given associations with type 2 diabetes as $\text{condFDR}(\text{fracture} | \text{T2D})$ and *vice versa*, and considered the significance cutoff < 0.01 . We used conjunctive FDR (conjFDR) to identify SNPs jointly associated with type 2 diabetes and fracture. After repeating the condFDR procedure for both traits, the conjFDR analysis reported the loci that exceed a condFDR significance threshold for two traits simultaneously (the maximum between the condFDRs for both traits), $\text{conjFDR} < 0.05$ was set as the significance.

We employed the Summary-data-based Mendelian Randomization (SMR) method developed by colleagues (Zhu et al., 2016 [DOI](#)) to test the association of the expression level of gene *RSPO3* with BMI (summary data available from Locke et al (Locke et al., 2015 [DOI](#))), waist circumference and waist-hip ratio (summary data available from Shungin et al (Shungin et al., 2015 [DOI](#))), type 2 diabetes and fracture using summary-level data from GWAS and expression quantitative trait loci (eQTL) data of the subcutaneous adipose tissue (9,962,255 SNPs included) from the GTEx database (release v8) (<https://www.gtexportal.org/home/> [DOI](#)) (Consortium et al., 2017 [DOI](#)). In the SMR analysis, the top cis-eQTL genetic variants were used as the instrumental variables (IVs) for gene expression. Additionally, we downloaded the cis-pQTL summary data for the circulating *RSPO3* reported in the study by Sun et al (Sun et al., 2018 [DOI](#)), and performed MR analyses to determine the association of circulating *RSPO3* with type 2 diabetes and fracture risk using the inverse variance-weighted (IVW) MR approach.

Observational analyses

For the 439,982 UK biobank samples (see foregoing description of study participants), we only focused the participants diagnosed with T2D within the 10-year period from 1 January 2006 to 31 December 2015, leaving 425,772 participants (with 14,860 type 2 diabetes patients) (Supplementary Fig. 1). Here, each type 2 diabetes patient had a diagnosis date (i.e., the reference date), we first calculated the onset age, then among the participants who were free of T2D, we selected 27 participants (whenever possible) whose age at the reference date (± 3 years) could be matching to the onset age as controls. In total, 363,884 controls were individually matched with 6-year age band at the reference date. We prospectively followed these type 2 diabetes patients and controls from the reference date until diagnosis of fracture, death, emigration, 19 April 2021 (diagnose a fracture of the last person in the cohort), whichever came first. Survival time was calculated based on whether the patient had a fracture. If individuals had a fracture, the survival time is calculated as the time of the first diagnosis of fracture minus the reference date. If individuals did not have a fracture, it was defined as the minimum time of the reference date to

diagnose a fracture of the last person in the cohort (19 April 2021), death, or emigration date. We excluded 25,865 participants with fracture diagnosis and non-fracture happened death or emigration before the reference date, leaving 352,879 participants included in the final analysis (Supplementary Fig. 1). Cox proportional hazards regression, as a statistical method to analyze the effect of risk factor on the time it takes for a specific event to happen, was used to test the relationship between T2D and fracture. Meanwhile, multiple linear regression analyses were performed to test the association between T2D and BMD.

First, we adjusted for clinical risk factors including reference age, sex, BMI, physical activity, fall history, HbA1c and medication treatments to examine the relationship between T2D and fracture. To examine the intermediary effect of risk factors on the relationship between T2D and fracture, the mediation analysis was performed using the R packages of “mediation”. Individuals treated with any glucose-lowering medication including insulin product, metformin, troglitazone, pioglitazone, rosiglitazone, tolbutamide, glibenclamide, gliclazide, glipizide, gliquidone, glimepiride, chlorpropamide, tolbutamide, repaglinide and nateglinide were recorded as having received medical treatment. Second, we defined 5 risk factors: $\text{BMI} \leq 25 \text{ kg/m}^2$, no physical activity, falls in the last year, $\text{HbA1c} \geq 47.5 \text{ mmol/mol}$ and antidiabetic medication treatments. Based on the number of risk factors, we grouped 14,860 individuals with T2D into subgroup for analysis. This analysis was adjusted for reference age, sex. For the BMD analysis, the age when attended assessment center was included in the analysis instead of reference age.

Data Availability

All data produced in the present work are contained in the manuscript

Article and author information

Authors' contributions

H-F.Z. conceptualized and designed the study. P-P.Z. conducted the analyses and drafted the paper. Z-F.S., L.X., P.L., W-J.X., C-D.Y., Z-W.X., M-Y.Y., Y.Q., J-D.Z., J-X.G., and D.K. helped in the data acquisition, result interpretation, and manuscript editing. The authors read and approved the final manuscript.

Competing interests

The authors declare that they have no competing interests.

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Ethics

All individuals provided written informed consent. The North West Multi-Centre Research Ethics Committee approved the UK Biobank ethical application (reference number: 16/NW/0274).

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Reviewer #1 (Public Review):

Summary:

The manuscript of Zhao et al. aimed at investigating the relationships between type 2 diabetes, bone mineral density (BMD) and fracture risk using Mendelian Randomization (MR) approach.

The authors found that genetically predicted T2D was associated with higher BMD and lower risk of fracture, and suggested a mediated effect of RSPO3 level. Moreover, when stratified by the risk factors secondary to T2D, they observed that the effect of T2D on the risk of fracture decreased when the number of risk factors secondary to T2D decreased.

Strengths:

- Important question
- Manuscript is overall clear and well-written
- MR analyses have been conducted properly, which include the usage of various MR methods and sensitivity analyses, and likely meet the criteria of the MR-strobe checklist to report MR results.

Weaknesses:

- Previous MR studies on that topic have not been discussed
- Multivariable MR could have been used to better assess the mediative effect of BMI or RSPO3 on the relationships between T2D and fracture risk.

- <https://doi.org/10.7554/eLife.89281.1.sa1>

Reviewer #2 (Public Review):

The authors employed the Mendelian Randomization method to analyze the association between type 2 diabetes (T2D) and fracture using the UK Biobank data. They found that "genetically predicted T2D was associated with higher BMD and lower risk of fracture". Additionally, they identified 10 loci that were associated with both T2D and fracture risk, with the SNP rs4580892 showing the highest signal. While the negative relationship between T2D and fracture has been previously observed, the discovery of these 10 loci adds an intriguing dimension to the findings, although the clinical implications remain uncertain.

- <https://doi.org/10.7554/eLife.89281.1.sa0>